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Education

H.B.Sc., Mathematics and Statistics 2018-2022
University of Toronto, Toronto, ON

Technical Skills

Data: Python, Numpy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Seaborn, SQL, NoSQL, R, Excel
Other: shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, Astropy, Selenium, NLTK, $\text{\LaTeX}$, regular expressions

Relevant Coursework

2019 Fall	Stars and Planets
2019 Fall	Probability and Statistics I
2019 Fall	Ordinary Differential Equations
2019 Summer	Multivariable Calculus
2019 Winter	Introduction to Computer Science
2018-2019	Calculus
2018-2019	Linear Algebra I, II
2018 Fall	An Introduction to Statistical Reasoning and Data Science

Independent Coursework

2019	Data-Driven Astronomy , The University of Sydney (Coursera) Pulsars, black holes, exoplanets, stellar evolution, galaxies. Image stacking, crossmatching, data management, regression, classification. Implementation in Python with Astropy, and SQL.
2019	Intro to Deep Learning with PyTorch , Udacity Mathematics of neural networks. Data preparation. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.
2018	Intro to Machine Learning , Udacity Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with sklearn.

Projects

2019	Image classification with CNNs Cleaned a dataset of several thousands of images of flowers. Optimized a deep learning model in Python and applied cutting-edge techniques to achieve 99% accuracy in the classification of species of flowers. Could potentially be adapted for classification of astronomical objects.
2018	Analysis of global Internet usage statistics Spearheaded the programming aspect of a collaborative statistical analysis project. Proposed research questions and used R to develop data-backed answers. Gave a presentation of results.

Volunteering

2015-2018	Python Workshop Lead , Sir Winston Churchill Secondary, Vancouver, B.C. Coordinated and delivered workshops to prepare students for a national programming contest.
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Interests

Languages: English (native), French, Chinese, Norwegian (conversational)
Other: *nix customization, urban beekeeping, baking, reading, computer programming